

Differential Expression

- Assignment #2 is due on Tue (2/17) at 11:59pm
- The next assignment is a 1-page summary of your proposed project (see next), due on March 3rd
- After a today's lecture, free computer time to work on any past exercises

Guidelines for Choosing a Project

- You will likely get the most out of this project if it relates to your own research. If you already have data in hand or will soon, I encourage you to use it.
- If you do not have your own data, I would recommend using the same type of data you will at some point analyze for your particular organism, which will make it easy to adapt the analysis to your specific data.
- If you have no data and no longer-term research project in mind, there are lots of options. If you have a research mentor, talk with them if you are stumped over a project to pursue. If not, meet with me to discuss.

Guidelines for Choosing a Project

- Carefully define the problems you wish to address or the questions you wish to answer
- They can be biological (e.g., I want to identify top candidate genes that differ in expression between pesticide resistant and pesticide sensitive plants)
- Or methodological (e.g., My model organism does not have a perfect reference genome, so I want to compare the counts obtained after using three different aligners)
- Define the scope of your project to fit the timeline. You'll need a preliminary analysis on April 2nd, so don't be overly ambitious. On the other hand, once you have a script working the analyses go fast. Incorporating 2-3 different techniques covered in class seems reasonable. And feel free to work with your research mentor on the most pressing analyses.

Project Planning Checklist...

- What sequence data will you use and how will you obtain it? What format will the data be in (e.g., raw FASTQ)?
- Are there reference genomes you will need, and where can you find those (Ensembl, NCBI, organism-specific database)?
- If your project includes mapping reads to genes, where will you obtain the gene annotations?
- What other types of annotation data will you need (e.g., GO annotations)?
- What programs do you plan on using for analysis?

The One Page Project Proposal Summary

- One paragraph of background or introduction to your project
- Next paragraph is the specific aims or questions that you would like to address, and how you are going to address those (you do not need to have a firm understanding of what programs you will use, but you should have some analysis goals in mind)
- It's probably helpful to discuss this with your advisor and/or me ahead of time (me especially if you do not have a research advisor)

Differential Expression Analysis

Overall goal is to determine whether there are statistically significant and biologically meaningful differences in gene expression between samples

The overall principle is the same whether you're looking at transcriptomics (RNA-seq) or proteomics:

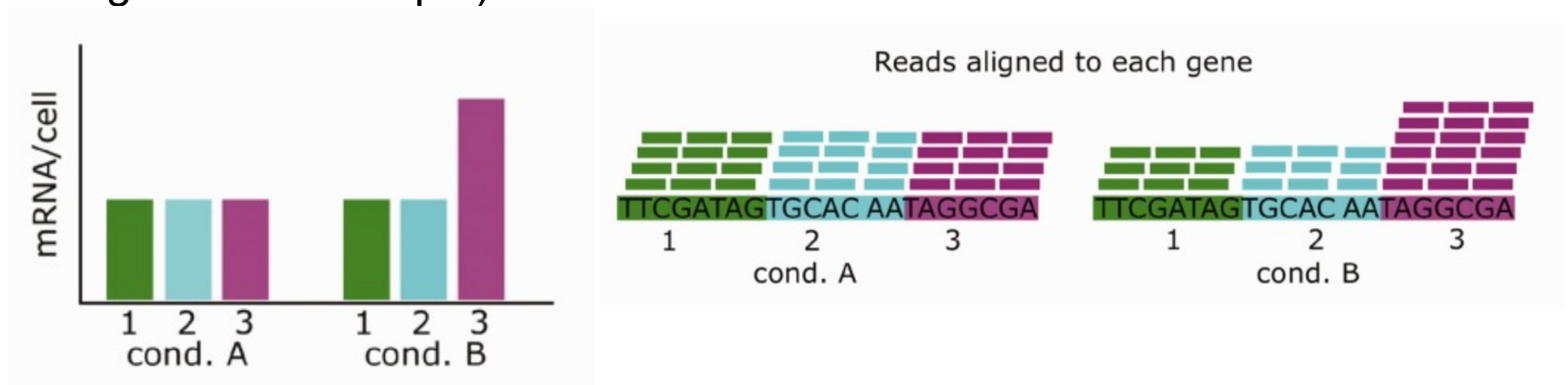
- 1) Need to “normalize” the data between samples to control for technical variation and bias
- 2) Need to apply a statistical test to determine whether differences in expression across comparisons are significant

Why does RNA-seq data need to be “normalized” ?

raw count data are not directly comparable because...

sequencing depth differs between samples (e.g., 20 million reads in sample A, 30 million in sample B, etc.)

the proportion of mRNA corresponding to a given gene may change across conditions (e.g., the total number of “available” counts decreases for the majority of genes if there are a few highly expressed genes in a sample)



transcripts from different genes have different lengths

Main types of normalization

- 1) Within a sample
- 2) Within a dataset (between samples)
- 3) Across datasets (between samples and between experiments)

Within sample normalization

- 1) Normalize to sequencing depth per sample
 - common solution is to divide by the number of reads in the sample and multiply by 1 million (counts per million or cpm)
- 2) Normalize to read length
 - longer transcripts will have proportionally more reads mapped to them (e.g., at equal expression levels, a 2-kb gene will have twice the mapped reads as a 1-kb gene)
 - divide reads by read length (usually in kilobases, so reads per kb)

Reads/Fragments per Kilobase Million (R/FPKM):

- 1) Count up the total reads in a sample and divide that number by 1,000,000 – this is the “per million” scaling factor
- 2) Divide the read counts by the “per million” scaling factor. This normalizes for sequencing depth, giving you reads per million (RPM)
- 3) Divide the RPM values by the length of the gene, in kilobases. This gives you RPKM.
- 4) FPKM takes into account paired-end sequencing, so accounts for the fact that two reads (forward and reverse) map to one fragment. Dividing by two if both reads were mapped (so as to not double count) keeps the numbers identical to single-end RPKM values.

Transcripts Per Million (TPM)

Minor difference compared to FPKM: first normalizes to gene length and then normalizes for sequencing depth

- 1) Divide the read counts by the length of each gene in kilobases. This gives you reads per kilobase (RPK).
- 2) Count up all the RPK values in a sample and divide this number by 1,000,000. This is your “per million” scaling factor.
- 3) Divide the RPK values by the “per million” scaling factor. This gives you TPM.

TPM vs R/FPKM

$$\text{RPKM} = 10^9 * \frac{\text{Reads mapped to the transcript}}{\text{Total reads} * \text{Transcript length}}$$

$$\text{TPM} = 10^6 * \frac{\text{reads mapped to transcript} / \text{transcript length}}{\text{Sum}(\text{reads mapped to transcript} / \text{transcript length})}$$

$$\text{TPM} = 10^6 * \frac{\text{RPKM}}{\text{Sum}(\text{RPKM})}$$

TPM vs FPKM

The minor difference in order of operations (TPM normalizes to gene length first and then sequencing depth) has one major implication:

All of the TPMs across samples always have the same sum,
which is NOT true of FPKM!

Thus, TPM is better for comparing relative abundance because you know the denominator is always the same

That said, comparing TPMs between samples is dangerous because the composition of mRNAs between samples are often different.

Improper Use of TPM

REVIEW

Misuse of RPKM or TPM normalization when comparing across samples and sequencing protocols

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In recent years, RNA-sequencing (RNA-seq) has emerged as a powerful technology for transcriptome profiling. For a given gene, the number of mapped reads is not only dependent on its expression level and gene length, but also the sequencing depth. To normalize these dependencies, RPKM (reads per kilobase of transcript per million reads mapped) and TPM (transcripts per million) are used to measure gene or transcript expression levels. A common misconception is that RPKM and TPM values are already normalized, and thus should be comparable across samples or RNA-seq projects. However, RPKM and TPM represent the relative abundance of a transcript among a population of sequenced transcripts, and therefore depend on the composition of the RNA population in a sample. Quite often, it is reasonable to assume that total RNA concentration and distributions are very close across compared samples. Nevertheless, the sequenced RNA repertoires may differ significantly under different experimental conditions and/or across sequencing protocols; thus, the proportion of gene expression is not directly comparable in such cases. In this review, we illustrate typical scenarios in which RPKM and TPM are misused, unintentionally, and hope to raise scientists' awareness of this issue when comparing them across samples or different sequencing protocols.

Improper Use of TPM

TPM (or FPKM) is often used to compare across samples

This is dangerous, because they are normalized within but not between samples

If your samples have different RNA compositions (e.g., huge changes in which transcripts are present or absent), the relative abundance is not comparable

This is why we do not calculate $\log_2$ fold changes from TPM, but instead from raw counts (where we can perform normalization across samples)

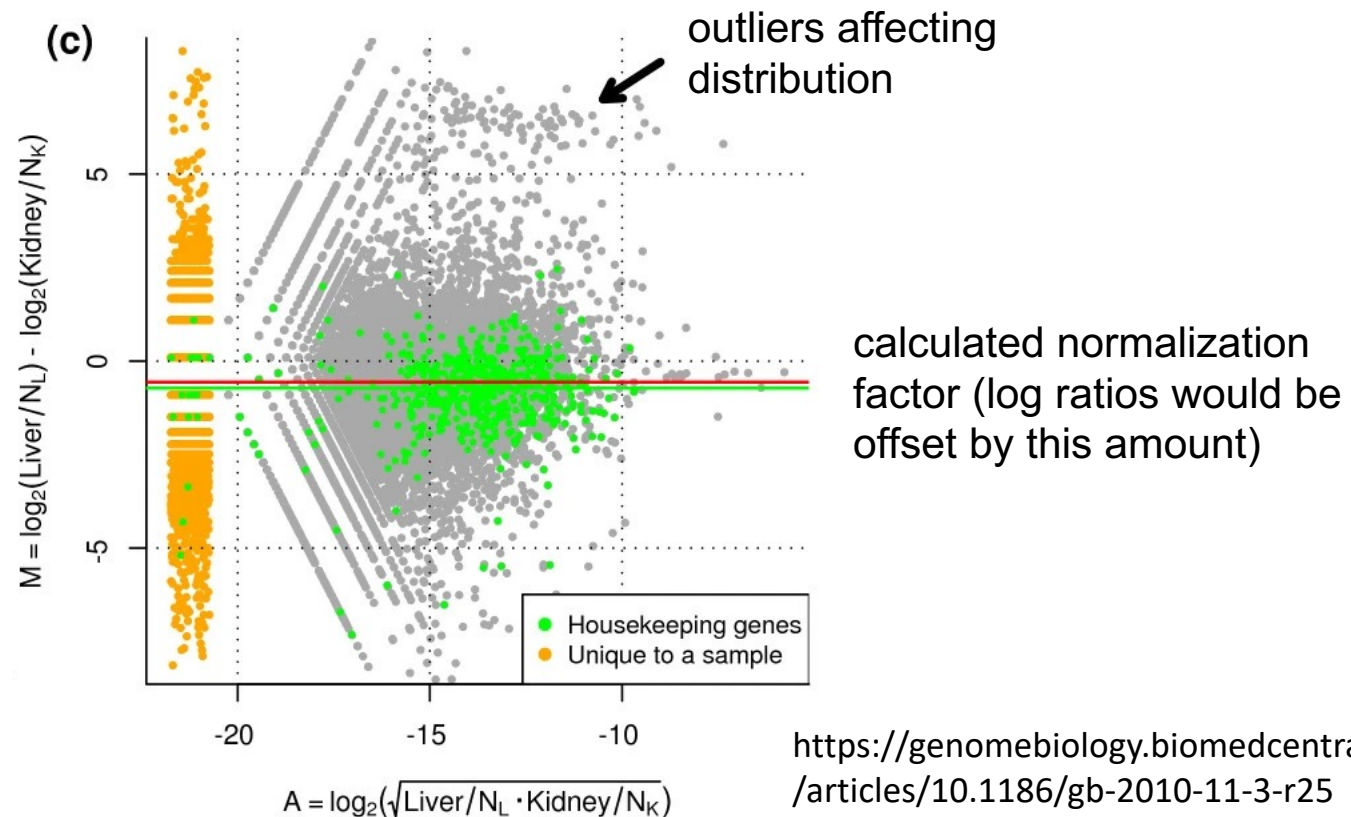
It is okay (and sometimes the only option) to use TPM for qualitative (not quantitative) comparisons, like Principal Component Analysis or clustering (which we'll talk about later)

Within dataset (between sample) normalization

Normalize to sequencing depth and composition:

edgeR: trimmed mean of M values (TMM)

- uses a weighted trimmed mean of the log expression ratios between samples (upper and lower outliers removed before averaging)



<https://genomebiology.biomedcentral.com/articles/10.1186/gb-2010-11-3-r25>

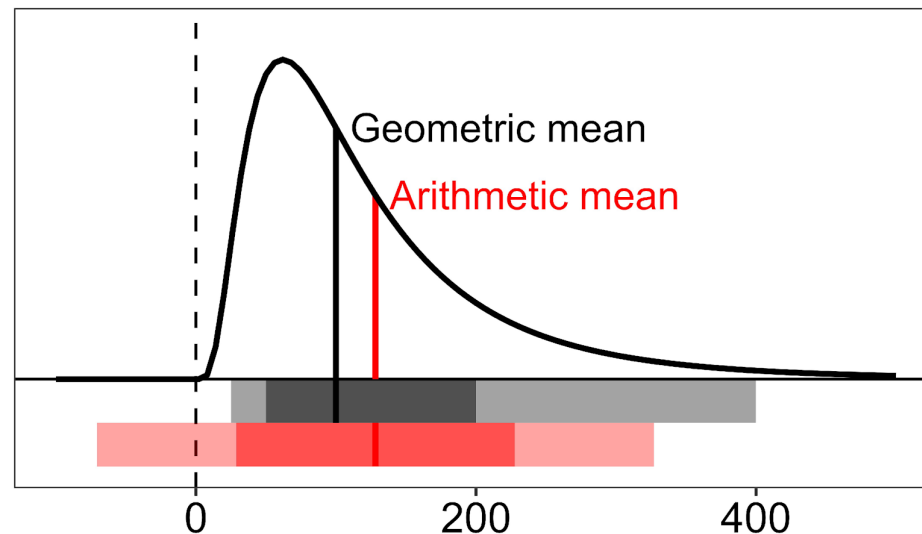
Within dataset (between sample) normalization

Normalize to sequencing depth and composition:

DESeq2: median of ratios

- counts divided by sample-specific size factors determined by median ratio of gene counts relative to geometric mean per gene
- the geometric mean is the n^{th} root of products of all measurements and a better measurement of central tendency than the arithmetic

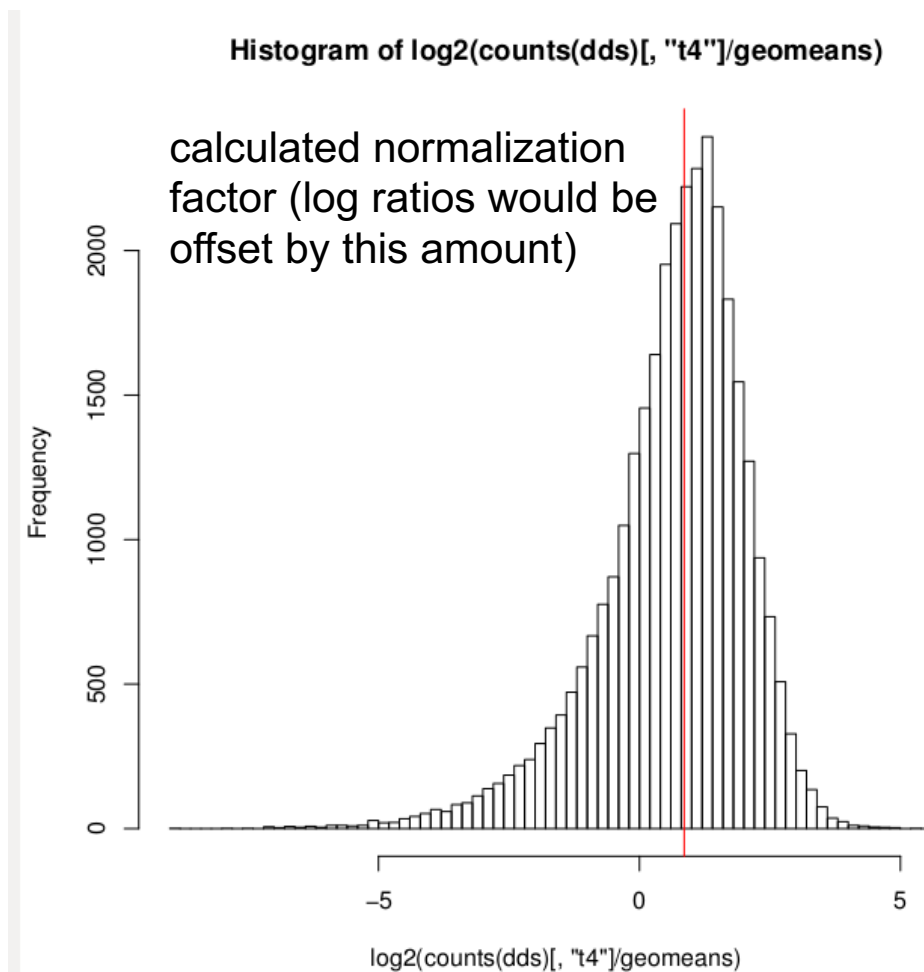
$$\text{geometric mean} = \sqrt[10]{73 * 65 * 70 * 71 * 68 * 90 * 74 * 67 * 89 * 73} = 73.6$$



Within dataset (between sample) normalization

Normalize to sequencing depth and composition:

DESeq2: median of ratios



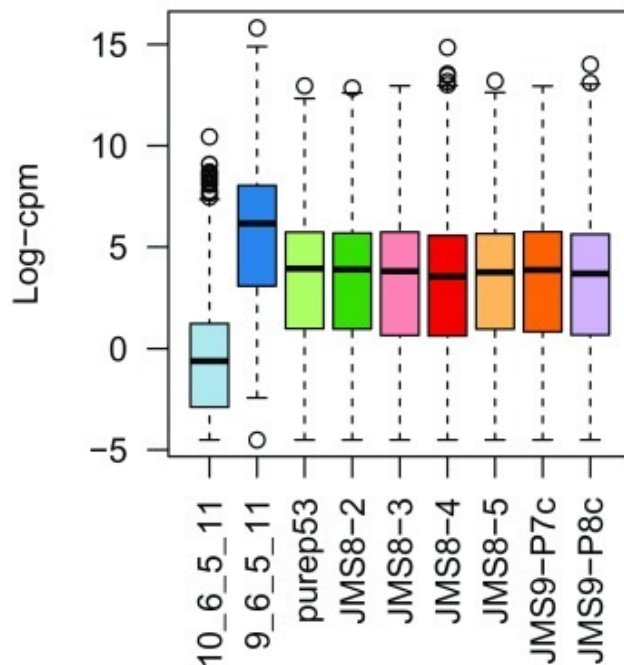
Within dataset (between sample) normalization

Normalize to sequencing depth and composition:

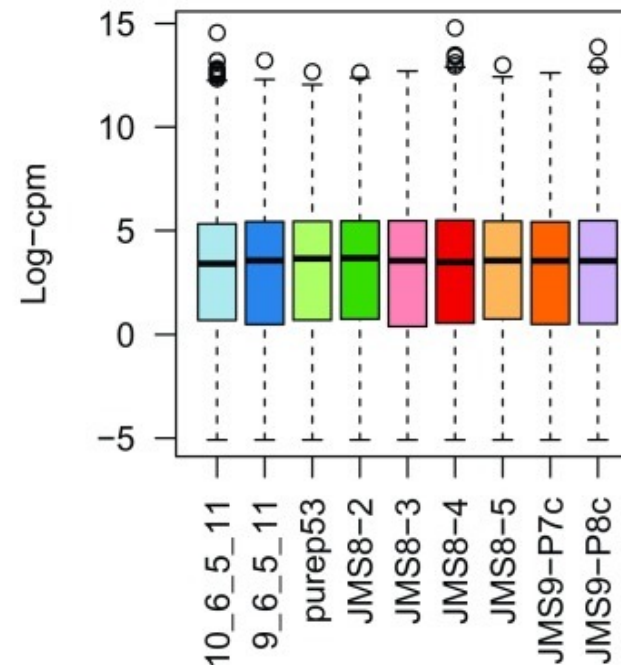
limma voom: log counts per million (logCPM)

- counts are normalized to sequencing depth and log transformed
- can apply TMM first (recommended)

A. Example: Unnormalised data



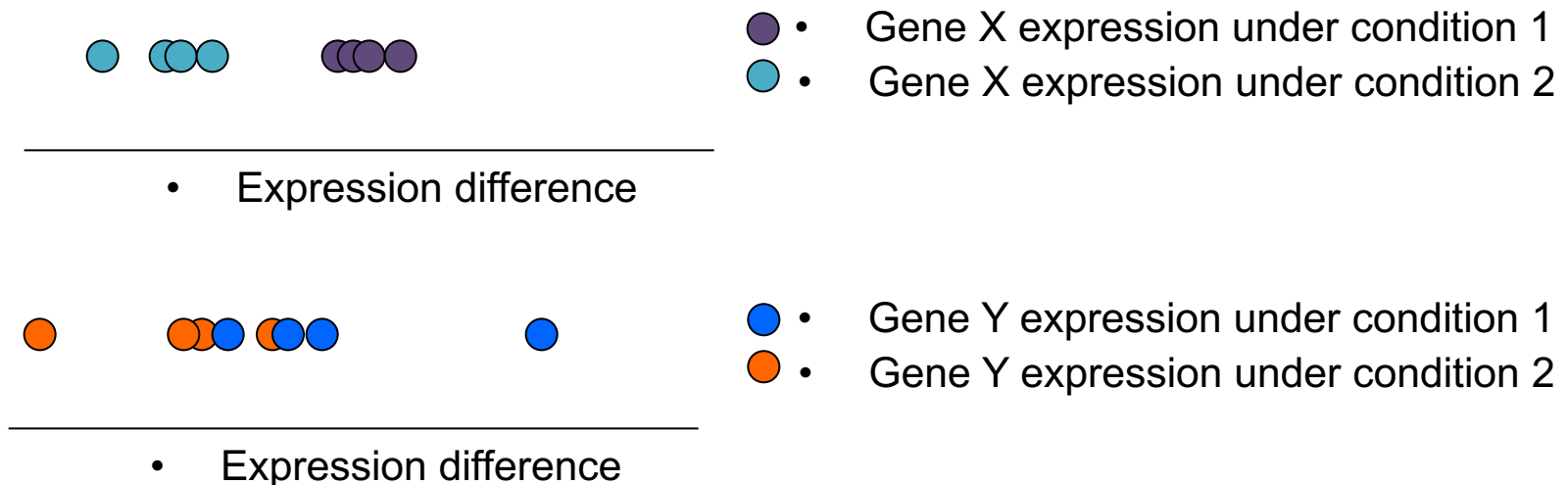
B. Example: Normalised data



Differential Expression

Methods of gene selection:

- 1) arbitrary fold-expression change cutoff (e.g., >2-fold difference in expression between samples)
- 2) statistically significant change in expression



How many biological replicates are needed in an RNA-seq experiment and which differential expression tool should you use?

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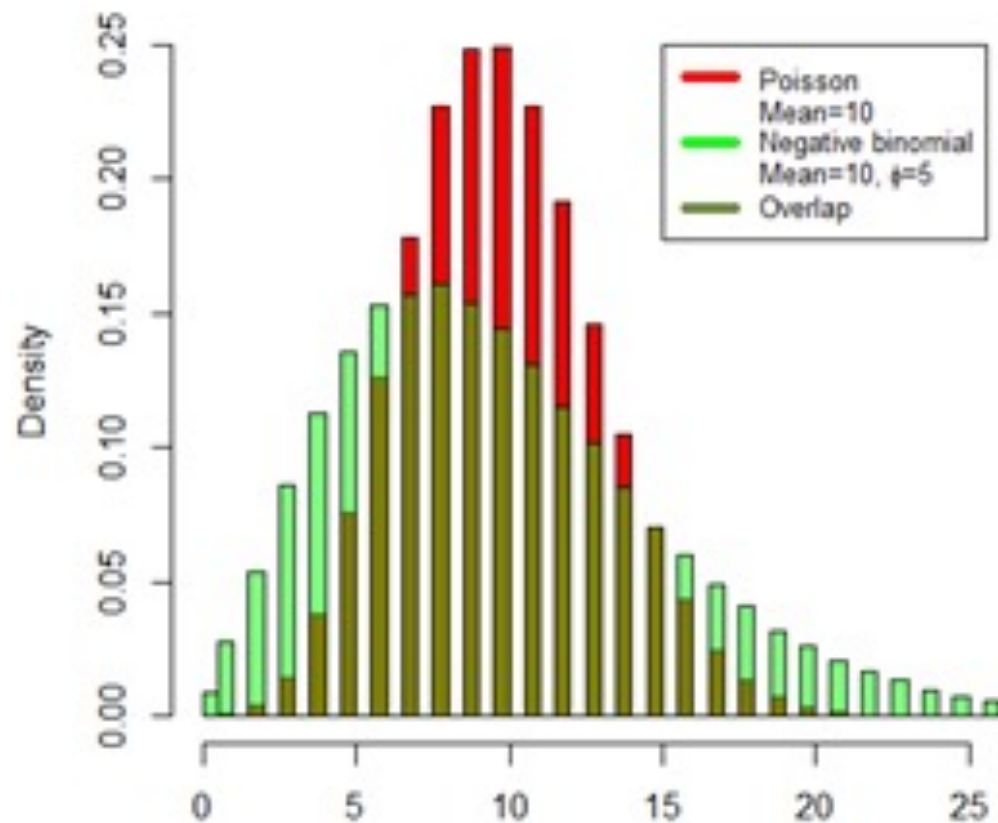
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- 48 biological replicates of yeast grown in two conditions
 - Provides “ground truth” for a *normal* experiment with 3-6 replicates
- For fewer than 12 replicates, edgeR and DESeq2 have comparable true positive and false positive performances
 - Both use “generalized linear models” based statistics to determine whether gene counts are significantly different between samples
 - Commonly-used outputs are log-fold changes and false discovery rate (FDR)-corrected p-values.

Overdispersion

RNA-seq counts do not follow a normal Poisson distribution, but instead a negative binomial distribution with long tails for reads with high counts

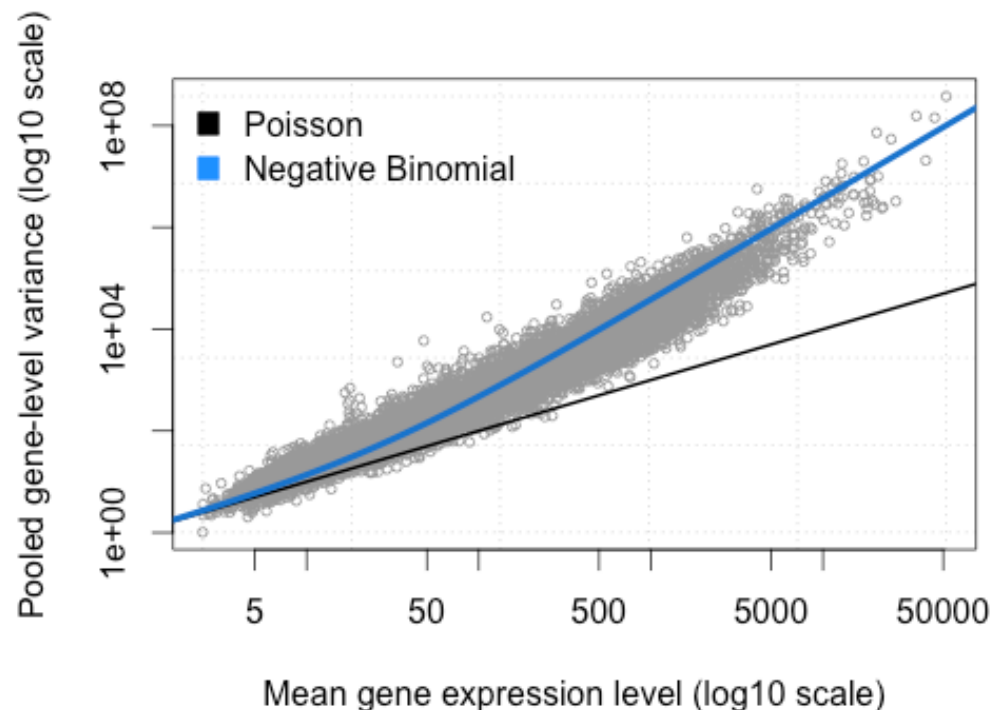


Overdispersion

For a normal Poisson distribution, the variance is equal to the mean

This is not the case for RNA-seq counts—highly expressed genes have higher variance (are overdispersed)

Best fit is to a negative binomial distribution (edgeR/DESeq2)



Calculating Significance

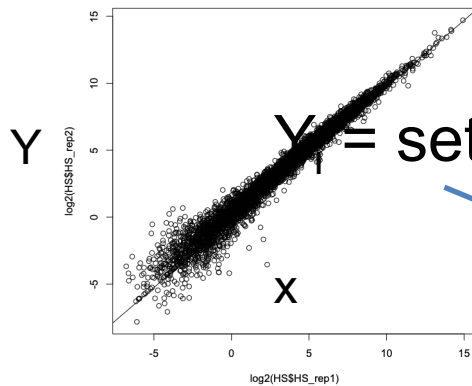
Can do a simple t-test or ANOVA for each gene, but that is generally underpowered

“classic” edgeR, DESeq2, and limma model the variance across all genes and use an “exact” test (similar to Fisher’s)

edgeR and limma also use generalized linear models, and edgeR uses quasi-likelihood tests that better control FDR based on uncertainty in the dispersion estimation

Limma and edgeR use linear models

Goal of linear model is to explain your set of data with a linear function



$$Y = mx + b$$

Y_i = set of normalized read values for Gene i

$$Y_i = \mu + R + S + D + S*D$$

Replicate
Effect

Strain
Effect

Treatment
Effect

Strain-by-Treatment
Effect

By including different factors that influence gene expression, we can more precisely quantify the contributions due to the factors we're interested in (e.g., treatment effect or maybe strain-by-treatment)

This allows us to generate p-values for the contribution of those specific factors